

Precept 4 Outline

ECO 202 TAs

September 30, 2026

1 Weekly Check In

1. TAs will have grading done for Exam 1 soon.
2. Any questions about the class, comments?
3. Next exam is about two weeks away.
4. Reminders about resources:
 - Come to office hours for any preceptor, as much as you want.
 - Dexin: Tuesdays, 4-5pm, tables outside of JRR A19
 - Emily: Thursdays, 4:30-5:30pm, JRR A74
 - Ryan: Fridays, 2-3pm, JRR A74
 - Come to office hours of the professor: Fridays, 3pm-4pm, JRR 278
 - Check out the McGraw Center:
 - Individual tutoring: [Link to page](#)
 - Group study with a tutor: [Link to page](#)
 - Reach out to us by email if you've tried the above but are seeking support beyond these things.

2 Precept practice problems

1. (Practice 6, Question 2)

A finite population contains four manufacturing firms $M_1, \dots, M_4$ and four service firms $S_1, \dots, S_4$. Compare two designs that select four firms.

- **Design A:** Select an SRS of four firms from all eight.
 - **Design B:** Make one fair SRS selection of two manufacturing firms and a separate fair SRS selection of two service firms, with all $6 \times 6 = 36$ cross-sector combinations equally likely.
- (a) Calculate the number of possible samples under Design A and the probability of each possible sample.
 - (b) Calculate the number of possible samples under Design B and the probability of each allowed sample.
 - (c) Calculate the inclusion probability of M_1 under each design. Explain why the same calculation applies to every firm.
 - (d) Find the probability of selecting the sample $\{M_1, M_2, S_1, S_2\}$ under each design.
 - (e) Find the probability of selecting $\{M_1, M_2, M_3, S_1\}$ under each design.
 - (f) Explain precisely why Design B is a probability sample and a stratified random sample but is not an SRS from the full eight-firm population, even though every firm has inclusion probability $1/2$.

2. (Practice 6, Question 3)

A fictional firm wants the mean weekly overtime hours during 2025 among everyone who worked for it at any point that year. The firm uses its December 31 current-employee roster as the frame. Former employees, many of whom report leaving because of long hours, averaged 12 overtime hours per week, while current employees averaged 5. Among current employees, people who worked more than 8 overtime hours were twice as likely to complete the survey as other selected employees. Supervisors collected the answers directly, and follow-up evidence suggests that respondents tended to underreport overtime in those conversations.

- For each stage—frame construction, response, and measurement—name the relevant bias mechanism and identify which units or values are disproportionately omitted or altered.
- Under the stated facts, determine the direction in which the frame mismatch moves the current-frame mean relative to the target-population mean.
- Determine the direction in which differential response moves the respondent mean relative to the current-frame mean.
- Determine the direction in which the stated response bias moves recorded reports relative to respondents' true overtime.
- Can the direction of the final reported mean relative to the target-population mean be determined from these facts alone? Explain why the signs of individual mechanisms do not supply their relative magnitudes.
- Explain why surveying every person on the current roster would remove one source of sampling variability but would not repair the target-population coverage or measurement problems.

3. (Practice 7, Question 2)

An outreach organization divides an eligible service area into four outreach zones, A–D. One experimental unit is one zone. The factor is the assigned outreach protocol, with levels **standard** and **enhanced**. The standard treatment uses the usual mailed notice; the enhanced treatment adds a follow-up phone call to the same notice. The outcome is the number of eligible households, out of five contacted in each zone, that schedule an enrollment appointment within one week. The following complete fictional table displays both potential outcomes only so that every causal quantity can be checked.

Unit	$Y_i(0)$	$Y_i(1)$
A	2	5
B	4	5
C	1	3
D	5	5

- Identify the units, factor, its two levels, the two complete treatments, the control, and the outcome with its units. Let $D_i = 1$ denote enhanced outreach and $D_i = 0$ denote standard outreach.
- Calculate every individual effect $Y_i(1) - Y_i(0)$ and the finite-sample average treatment effect

$$\text{ATE} = \frac{1}{4} \sum_{i=1}^4 [Y_i(1) - Y_i(0)].$$

- Under Assignment I, zones A and B receive enhanced outreach. Construct all four observed outcomes and calculate

$$\hat{\tau} = \bar{Y}_{D=1} - \bar{Y}_{D=0}.$$

- Under Assignment II, zones C and D receive enhanced outreach. Repeat the observed-outcome construction and calculate $\hat{\tau}$.

- (e) Explain why the potential outcomes and ATE are fixed across the two assignments while the observed outcomes and $\hat{\tau}$ change.
- (f) There are $\binom{4}{2} = 6$ equally likely assignments when exactly two of the four zones receive enhanced outreach. Enumerate the six possible enhanced-outreach pairs and calculate $\hat{\tau}$ for each. Verify that the estimates have mean 1.5 and range from -0.5 to 3.5 , then explain what each fact teaches about random assignment.

4. (Practice 7, Question 6)

A career center lets eligible applicants choose whether to attend a résumé workshop, then compares their earnings six months later. The attendees have higher mean earnings, and the center calls the difference the causal effect of the workshop.

- (a) Define the units, a sharply stated treatment or offer, D_i , $Y_i(1)$, $Y_i(0)$, and a finite-sample ATE.
- (b) Explain why the self-selected comparison does not by itself estimate that ATE. Name two pre-treatment variables that could affect both workshop choice and later earnings.
- (c) Repair the design using one of the following assignment mechanisms: complete randomization of a workshop offer, blocking by pre-treatment earnings, or matched pairs based on pre-treatment information. State the mechanism precisely, explain how the design provides replicated treated and control comparisons, and identify the causal contrast it supports.
- (d) State separately whether the applicant pool was obtained through probability sampling and what population the design directly describes.
- (e) Name two important limitations that could remain after valid random assignment.
- (f) Write a two-sentence conclusion template that an analyst could complete after observing the assignment-based mean difference, without claiming a receipt effect, an effect for every applicant, or automatic generalization.

3 Extra practice problems

1. In the context of sampling, what is a “parameter”?
2. What is a probability sample?
3. What, precisely, is a simple random sample?
4. Give an example of a “parameter” and of a “statistic”.
5. Suppose a population consists of 5 individuals, and we draw a Simple Random Sample of size 2. How many possible samples are there, and what is the probability that each is chosen as the sample?
6. Consider a population that consists of 5 individuals A, B, C, D, E. Construct a probability sample for a sample size $n = 3$ such that all individuals are equally likely to be in the sample, but the design is not a simple random sample.
7. Consider a simple random sample of size $n = 2$ from a population of 4 individuals A, B, C, D.
 - (a) What are the possible samples, and the associated probabilities?
 - (b) Is the event “A or B, or both, are in the sample” independent of the event “C is not in the sample”?
 - (c) Suppose we learn that A is in the sample. What is the conditional probability that also B is in the sample?
8. Consider a population of 4 individuals A, B, C, D. A, B are male, and C, D are female. Consider a stratified random sample of size $n = 2$, with strata defined by gender. What samples are possible, and what are their probabilities?
9. Consider a population of four individuals A, B, C, D. Describe a random sample design of size $n = 2$ so that each individual has the same chance to be in the sample, but the design is not a simple random sample. Explain.
10. A population consists of 5 individuals A, B, C, D, E. Design a probability sample of size $n = 3$ such that each individual is selected with the same probability, but the sample design is not a simple random sample.
11. Consider a population of five individuals A, B, C, D, E. Suppose we take a simple random sample of size $n = 4$. What is the probability that both A and B are in the sample?
12. Consider the first paragraph of the Class of 2025 pre-read *Moving Up Without Losing Your Way* below. In particular, do you think the literal interpretation of the sentence “If you had been born in San Jose, California, instead, your chances would have increased threefold” is warranted? Discuss using relevant jargon, and provide a reformulation of the sentence that is not subject to the same criticism.

Each of us starts off in a particular position in society. That position is determined by where and when you are born, the socioeconomic and educational level of your parents, your race, and your gender, among other things. Social scientists have produced vast amounts of evidence showing that these factors play a significant role in determining life prospects. In the United States, if you were born in the bottom fifth of the income scale in Charlotte, North Carolina, between 1980 and 1982, your chances of making it to the top fifth were 4.4 percent. If you had been born in San Jose, California, instead, your chances would have increased threefold.
13. Give an example of a “matched pair design”, and explain its appeal.
14. Observational studies suggest that countries with more generous unemployment benefits have higher rates of unemployment. If countries were to make unemployment benefits less generous, would this lead to lower unemployment rates? Discuss using relevant jargon.